Solutions
QA : Numbers - 04

CEX-Q-0222/25

1	2	**2**	4	**3**	3	**4**	1	**5**	1	**6**	4	**7**	4	**8**	1	**9**	–	**10**	–
11	1	**12**	4	**13**	–	**14**	1	**15**	3	**16**	2	**17**	2	**18**	4	**19**	1	**20**	2

1. 2 **Given:** 3798125P369 is divisible by 7.
Let us go by the options:
For P = 1, P = 7 and P = 9 the numbers 37981251369, 37981257369 and 37981259369 cannot be completely divided by 7.
When P = 6, the number 37981256369 divided by 7 gives 5425893767. There is no remainder.
Hence, correct option is (2).

2. 4 Let n = 9
Hence, 3n + 1 = 3 × 9 + 1 = 28 will be divisible by 4.

3. 3 Sum of the digits at the even places of the number N is equal to 9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45.
Sum of the digits at the odd places of the number N is equal to 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45.
Difference = 45 – 45 = 0.
Therefore, the number N is divisible by 11.
Hence, the remainder when (N + 1)(N + 2)… (N + 6)(N + 7) is divided by 11 is the remainder when 1 × 2 × 3 × 4 × 5 × 6 × 7 is divided by 11, i.e., 2.

4. 1 To find the remainder when a number N is divided by 3, we need to find the remainder when the sum of the digits of N is divided by 3. Since the sum of the digits of A is 13, the number A when divided by 3 leaves a remainder of 1.
Hence, $A^4 = (3x + 1)^4$ leaves a remainder 1 when divided by 3.

5. 1 Since 899 is divisible by 29, you can directly divide the remainder of 64 by 29, so 64 /29 will give 6 as a remainder, so option (1) is the answer.

6. 4 Suppose $x = 6k_1 + 4$ and $y = 6k_2 + 5$
$x^2 + y^2 = (6k_1 + 4)^2 + (6k_2 + 5)^2$
$= 36k_1^2 + 48k_1 + 16 + 36k_2^2 + 60k_2 + 25$
$= 36k_1^2 + 48k_1 + 36k_2^2 + 60k_2 + 41$
Obviously, when this is divided by 6, the remainder will be 5.

Short cut:

$r_1^1 + r_2^2$ will also give the same remainder when divided by 6 (where 'r_1' and 'r_2' are the remainders when 'x' and 'y' are divided by 6 respectively).

$\therefore \frac{4^2 + 5^2}{6} = \frac{41}{6}$ will give 5 as a remainder.

7. 4 Note that the difference between the divisors and the remainders is constant.
2 – 1 = 3 – 2 = 4 – 3 = 5 – 4 = 6 – 5 = 1
In such a case, the required number will always be [a multiple of LCM of (2, 3, 4, 5, 6) – (The constant difference)].
LCM of (2, 3, 4, 5, 6) = 60
Hence, the required number will be 60n – 1.
Thus, we can see that the smallest such number is (60 × 1) – 1 = 59
The second smallest is (60 × 2) – 1 = 119
The third smallest is (60 × 3) – 1 = 179
So between 1 and 200, there is only three such number, viz. 59, 119 and 179.

8. 1 Successive division means we first divide by the divisor, then the quotient is divided by the next divisor and then this quotient is divided by the next divisor and so on. Remainders are obtained at each of these steps.
Let N is the number which when successively divided by 3, 4 and 7 leaves remainders as 2, 1 and 4 respectively. So N divided by 3 leaves a remainder of 2. Let the quotient is k. This quotient k when divided by 4 leaves a remainder of 1 and a quotient of m. This quotient m when divided by 7 leaves a remainder of 4 and a quotient of n.
So m = 7n + 4 and k = 4m + 1 = 4(7n + 4) + 1 and N = 3k + 2

Or

N = 3(4(7n + 4) + 1) + 2 = 84n + 53 = 84n + 42 + 11 = 42(2n + 1) + 11.
So the remainder when N is divided by 42 is 11.

9. Let the remainder is R. This means 67516 – R and 68488 – R are divisible by that 3 digit number. If two numbers are divisible by a particular number, then their difference should also be divisible by that number. So, 68488 – R – 67516 + R = 972. So all factors of 972 will leave the same remainder by dividing 68488 and 67516. $972 = 2^2 \times 3^5$. The factors of 972 are 1, 2, 3, 4, 6, 9 and so on. Similarly factors of 972 which are of three digits are 972, 486, 324, 243, 162 and 108. Hence 6 values are there.

10. The number is of the type 7k + 2 and is also of the type $8K_1 + 4$. Hence
$7k + 2 = 8k_1 + 4$, where k and k_1 are natural numbers.

$7k = 8k_1 + 2$ Or $k = \frac{(8k_1 + 2)}{7}$

We get natural number value of k by putting $k_1 = 5$.
So the smallest number that satisfies the given condition = 8 × 5 + 4 = 44
The numbers that satisfy the given condition is given by k × LCM (7, 8) + 44, where k is a whole number.
In order to find the smallest three digit number that satisfies the given condition, k has to be 1.
Hence, the required number = 1 × 56 + 44 = 100.

11. 1 We can write $(37)^{47}$ as $(38 - 1)^{47}$ which when divided by 19 leaves a remainder of $(-1)^{47}$ or 18.

12. 4 $\text{Rem}\left(\frac{4^1}{6}\right) = 4$

$\text{Rem}\left(\frac{4^2}{6}\right) = 4$

$\text{Rem}\left(\frac{4^3}{6}\right) = 4$

$\text{Rem}\left(\frac{4^4}{6}\right) = 4$

Hence, any power of 4 when divided by 6 leaves a remainder of 4.

13. $\text{Rem}\left[\frac{5^{4k}}{6}\right] = \text{Rem}\left[\frac{(-1)^{4k}}{6}\right] = \text{Rem}\left[\frac{1}{6}\right] = 1.$

14. 1 The terms $1^3, 5^3, 9^3, ..., 21^3$ are of the form 4k + 1.
The terms $3^3, 7^3, 11^3, ..., 23^2$ are of the form 4k + 3.

$= \text{Rem}\left[\frac{6 \times (4k+1) + 6 \times (4k+3)}{4}\right]$

$= \text{Rem}\left[\frac{6 \times 1 + 6 \times 3}{4}\right] = 0.$

15. 3 $15^{23} = (19 - 4)^{23} = 19x + (-4)^{23}$ where x is a natural number.
$23^{23} = (19 + 4)^{23} = 19y + (+4)^{23}$ where y is a natural number.
$15^{23} + 23^{23} = 19 (x + y) + 4^{23} + (-4)^{23} = 19 (x + y)$
The above expression is multiple of 19. Hence, the required remainder is 0.

16. 2 Given expression is
n! + (n! + 1) + (n! – 2) + (n! + 3) ..... + (n! – 2006)
= 2007 × n! + (1 – 2 + 3 – 4 ..... – 2006)
= 2007 n! + 1003 × (–1)
Now, for n = 1003, the expression 2007 n! – 1003 is clearly divisible by 1003.
Hence, the remainder is zero. Option (2) is the correct choice.

17. 2 $7^{6n} - 6^{6n}$
Putting n = 1
$7^6 - 6^6 = (7^3 - 6^3)(7^3 + 6^3)$
This is multiple of $7^3 - 6^3 = 127$ and $7^3 + 6^3 = 559$ and 7 + 6 = 13.
Hence, $7^{6n} - 6^{6n}$ is divisible by all of above except 81.

18. 4 $a^n - b^n$ (n > 0 and integer) is always divisible by (a – b).
$\therefore 6^{6n} - 5^{6n} = (6^2)^{3n} - (5^2)^{3n} = 36^{3n} - 25^{3n}$
∴ This is divisible by (36 – 25) = 11
Similarly, $6^{6n} - 5^{6n} = (6^3)^{2n} - (5^3)^{2n} = 216^{2n} - 125^{2n}$
∴ This is divisible by (216 – 125) = 91 and (216 + 125) = 341.
341 is divisible by 11, 91 and 341.

19. 1 $N \times M = (5^{2p} - 1)(5^q + 1)^2 (5^r + 1)^2$
$= (5^{2p} - 1)(5^{2q} + 1 + 2.5^q)(5^{2r} + 1 + 2.5^r)$
$(5^{2p} - 1)$ when divided by 6 will yield a remainder of '0'.
⇒ (N × M) when divided by 6, gives the remainder 0.

20. 2 If $Q = 2^P$, then the highest power of 2 in Q! is Q – 1.
$\therefore Q! = (2K + 1) \times 2^{Q-1}$

$\therefore \text{Rem}\left[\frac{Q!}{2^Q}\right]$

$= \text{Rem}\frac{(2K+1)2^{Q-1}}{2^Q} = 2^{Q-1} \times \text{Rem}\left(\frac{2K+1}{2}\right) = 2^{Q-1}.$